

Ray Simon L. Bantaculo

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Education

University of Science and Technology of Southern Philippines

Bachelor of Science in Computer Engineering

Thesis: TANIM: An Agricultural System with Integrated IoT Soil Health Monitoring Device and Crop Recommendation

Using LightGBM

Cagayan de Oro City, Philippines

April 2022 – Present

Experience

Ordlabs

Full Stack Developer Intern

Philippines (Hybrid)

January 2026 – March 2026

- Built and maintained web applications using React, Next.js, and RESTful APIs.
- Collaborated on database schema design, optimization, and backend service integration.
- Participated in Git-based version control, Agile stand-ups, and cross-functional task reviews.
- Assisted with system data entry operations and supported marketing campaigns for product rollouts.

Projects

TANIM: IoT Soil Health Monitoring & Crop Recommendation

Academic Project

ML & Backend Engineer

August 2025 – May 2026

- Trained and tuned the LightGBM classifier on regional soil datasets, achieving an 82.29% global accuracy for crop recommendations.
- Architected and deployed the API server infrastructure, structuring request/response routing for model inference and sensor data streams.
- Achieved a 100% UAT success rate and an “Excellent” System Usability Scale (SUS) score of 84.25 during field testing.

App Release Management System (ARMS)

Full Stack Intern Project

Backend Developer

January 2026 – March 2026

- Developed core RESTful API endpoints for application release management and version tracking using Express v5 and TypeScript.
- Designed MongoDB schemas and referential relations using Mongoose, implementing update/delete routes for nested sub-resources.
- Refactored backend analytics services and integrated Zod schemas for robust environment and request payload validation.

CPELEX: Smart RFID Tollgate System

Embedded Systems Project

Embedded Systems & Mobile Developer

December 2025

- Developed a cross-platform companion mobile app using Flutter to track real-time toll transactions and manage account balances.
- Programmed the Raspberry Pi in Python, configuring GPIO interfaces for ultrasonic sensors, servo gates, and user-feedback LEDs.
- Engineered an offline-first transaction queue saving RFID scans to local JSON, auto-syncing with Firebase upon connection.

Skills & Interests

Technical: Frontend/Web (HTML, CSS, JS, Next.js, Tailwind, TS), Backend (Node, Express, REST APIs, FastAPI, MongoDB, Firebase), Mobile (Flutter, Kotlin), Languages (Python, Java, Kotlin, Dart), IoT/Embedded (Arduino, ESP32, Raspberry Pi, sensors, motor control), Basic Networking.

Tools & Databases: Git, GitHub, VS Code, Android Studio, Firebase (Firestore, RTDB), MongoDB, SQL.

Certifications: CCNA: Switching, Routing, and Wireless Essentials • LaTeX Typesetting Workshop • Agricultural Training Institute (RTC X)

Soft Skills: Problem-solving, debugging mindset, fast learner, remote collaboration & communication.